

	sparse_attention A	sparse_attention B
Cube Matmul for both QK and PV		✓
Cube / Vector stages on separate streams		✓
Event-driven cross-chunk pipeline		✓
Gather KV once; reuse across QK and PV		✓
Explicit L1 depth / L0 double-buffer tiling	—	✓
TQue-managed producer / consumer buffers	—	✓
ReduceSum API instead of a manual sum tree	✓*	✓
Double buffering and input prefetch	—	✓
Vector arithmetic / UB reuse / mixed precision	✓*	✓